

Action-oriented, results-driven analyst with a strong foundation in statistical modeling, large-scale dataset analysis, Python, and SQL. Experienced in ensuring data accuracy, solving complex data issues, and delivering reliable results under tight deadlines in applied research settings. Seeking to apply these skills and grow as a data scientist in a business setting to generate high-quality insights that support data-driven decision making. Known for clear, proactive communication and effective collaboration across technical and non-technical teams.

SKILLS

Languages & Tools	Python (pandas, NumPy, scikit-learn), SQL, R, Git, MS Office Suite
Data & Analytics	Data validation and cleaning, pipeline automation, statistical analysis, regression modeling
Visualization	Tableau, matplotlib, seaborn, ggplot2
Other	Jira, Unix, Jupyter, Spanish (Elementary proficiency), Chinese (Elementary proficiency)

RELEVANT EXPERIENCE

Data Science Intern - CBB Analytics May 2026 — Present
Remote

- Developed data-driven narratives around NCAA Tournament performance and NBA Draft outcomes, using SQL, Python, and Tableau to transform multi-season player data into 5+ visualizations and dashboards which communicate novel statistical findings to non-technical audiences.
- Self-led research design and applied computational analytics to assemble, clean, join, and analyze large player datasets, finding meaningful trends and delivering actionable insights with SQL to grow a new company tool.

AI MOVE Fellow, Image Generation Domain Expert - Handshake Oct 2025 — Present
Remote

- Evaluated large language model (LLM) outputs for accuracy, consistency, and domain validity, applying structured review frameworks to improve model reliability and performance.
- Delivered 35+ individual projects with efficiency and adherence to updating guidelines in a fast-paced environment.

Graduate Researcher – Exo-Disk Modeling for SCALES at Keck Observatory Dec 2023 — Jun 2025
UC Irvine — Sallum Research Group

- Leveraged statistical models and novel data simulations in Python to evaluate signal detectability and performance predictions of next-generation technology.
- Delivered actionable insights from messy, high-volume datasets with immediate impact and collaborated cross-functionally to guide strategic decisions in a fast-paced research setting.

Teaching Assistant — UCI Physics Department Dec 2023 — Jun 2025

- Delivered simple, clear explanations of complex material to 500+ diverse learners across 7 physics and astronomy courses, sharpening client-facing communication, interviewing, and presentation skills.
- Organized structure for classes up to 150 students and completed outreach work, distilling astrophysics research in an easily understood manner.

Data Analyst – Dynamical Masses of Stellar Hosts (NASA-Funded Research) Aug 2020 — Aug 2024
Wesleyan University — Disk Detectives Research Group

- Led a NASA-funded astronomy research project from inception to final product using Python to analyze large datasets (100+ GB), applying automated regression, uncertainty quantification techniques, and modeling strategies to extract insights from 21 complex systems. Acted in a leadership role for younger peers.
- Developed reproducible Python-based pipelines for processing and visualizing high-dimensional, structured data, improving efficiency and accuracy in interpreting interferometric imaging outputs.

EDUCATION

Master of Science in Astronomy and Astrophysics, University of California, Irvine Sep 2025
GPA: 3.81/4.00 | Graduate coursework in machine learning, experimental research, climatology and communication skills.

Bachelor of Arts in Physics & Astronomy, Wesleyan University May 2023
GPA: 3.86/4.00 | Phi Beta Kappa | Honors Thesis | Minor in Data Analysis

CERTIFICATIONS

• Duke: Business Metrics for Data-Driven Companies (KPIs, Business Analytics, Dashboarding)	Oct 2025
• UMich: Introduction to Structured Query Language (SQL)	Sep 2025
• UCI: Accelerate-2-Industry Career Transition Certificate (Cross-functional Collaboration, Best Practices Leader)	Jun 2025
• AAS National Osterbrock Fellow	Sep 2024 — Jun 2025
• Certificate in Fellowship Mentorship and Writing Excellence	May 2024
• NA-ROAD Women and Girls in Astronomy Program – \$2,000 (sole proposer)	May 2024
• APS Women in Physics Group Grant – \$1,000 (1 of 3 contributing writers)	Mar 2024
• Mentoring Excellence Program Certificate	Oct 2023
• Phi Beta Kappa Honor Society	May 2023
• NASA CT Space Grant Consortium Undergraduate Scholarship & Research Grant – total \$8,000	March 2021 — May 2023

SELECTED PROJECTS

March Madness 2026 Bracket Prediction Model	Mar 2026
• Built a predictive ML modeling pipeline using a random forest algorithm to forecast game outcomes in the 2026 NCAA D1 Men’s Basketball Tournament. Aggregated and cleaned large-scale team datasets (e.g., efficiency ratings) going back to 2008.	
• Implemented feature engineering and evaluated model performance using cross-validation and probabilistic metrics (e.g., log loss, accuracy).	
• Generated bracket predictions with quantified probabilities, achieving 75% accuracy after hyperparameter tuning and finishing in the 99.4th percentile of all brackets on ESPN.	
NBA 2025-26 Rookie of the Year Model	Dec 2025
• Developed a supervised machine learning model using a random forest algorithm in Python to predict the 2025-26 NBA Rookie of the Year winner based on historical NBA data going back to 2010.	
• Achieved strong predictive performance (e.g., MAE: 0.0196) and identified the most influential features driving award outcomes.	